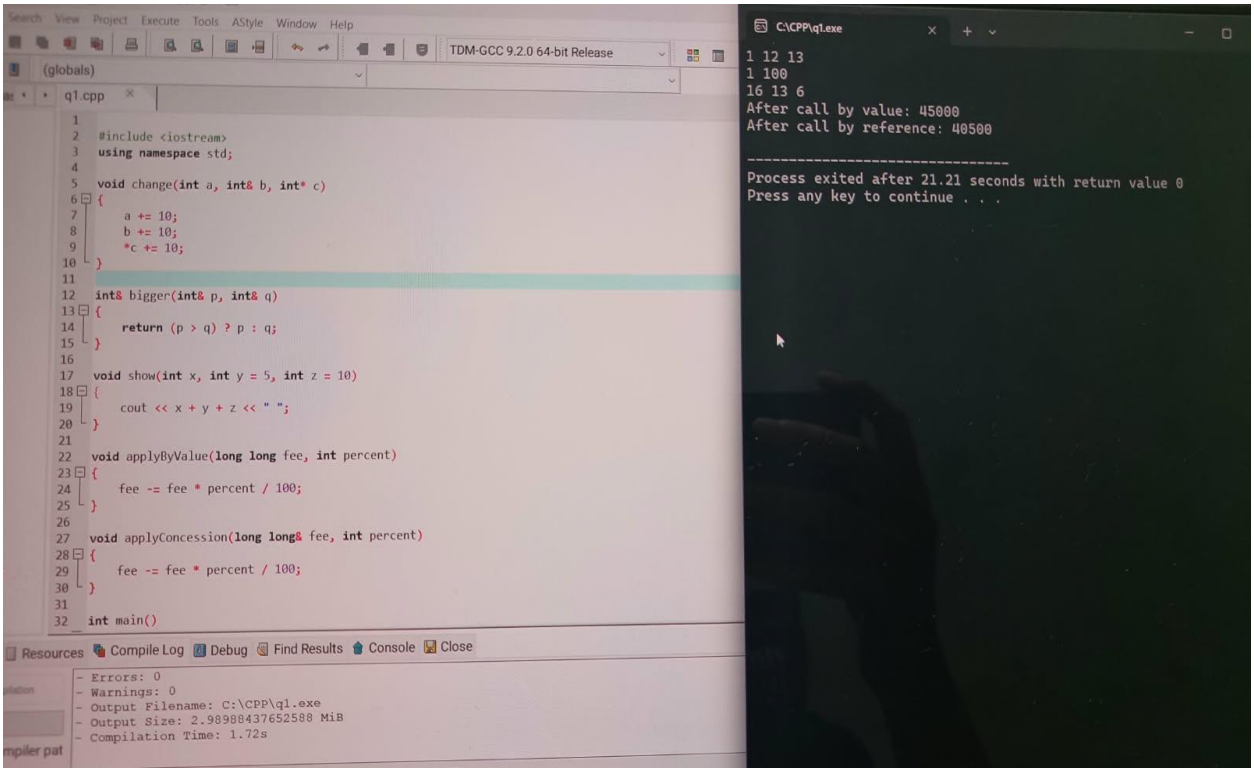
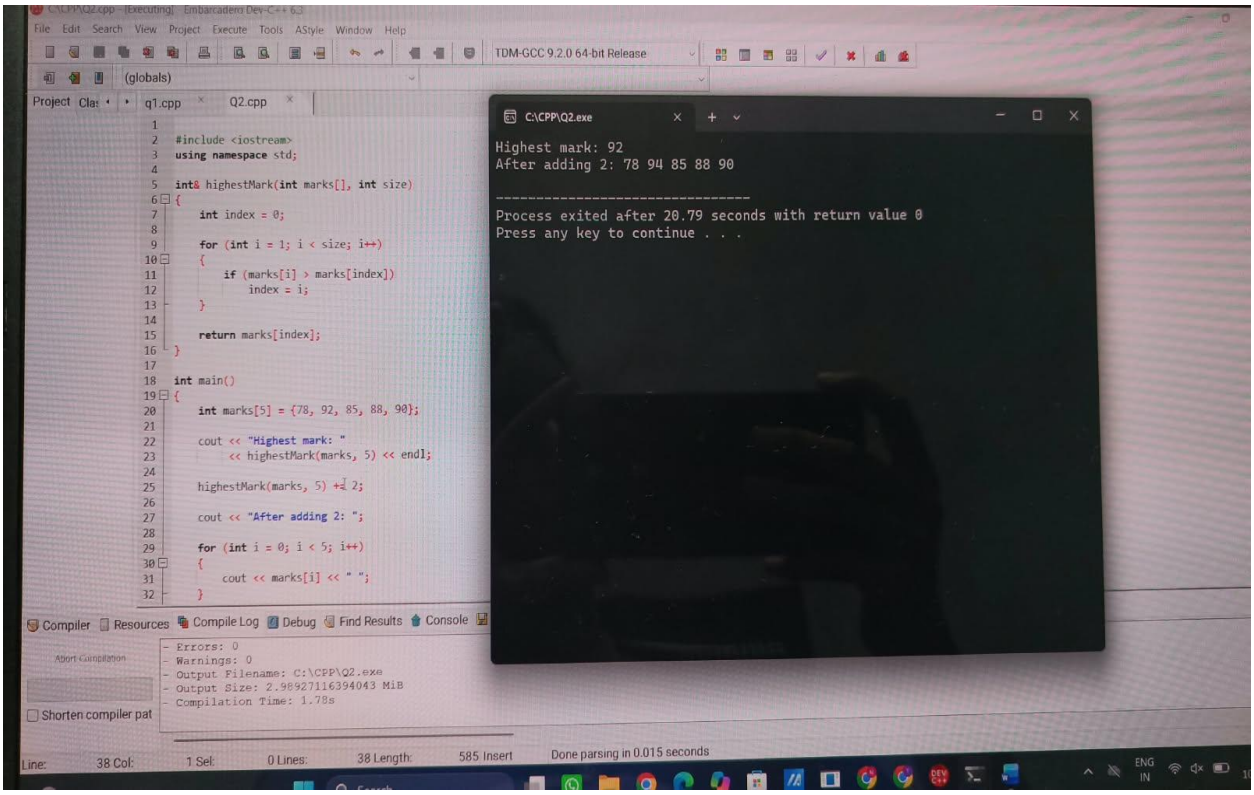
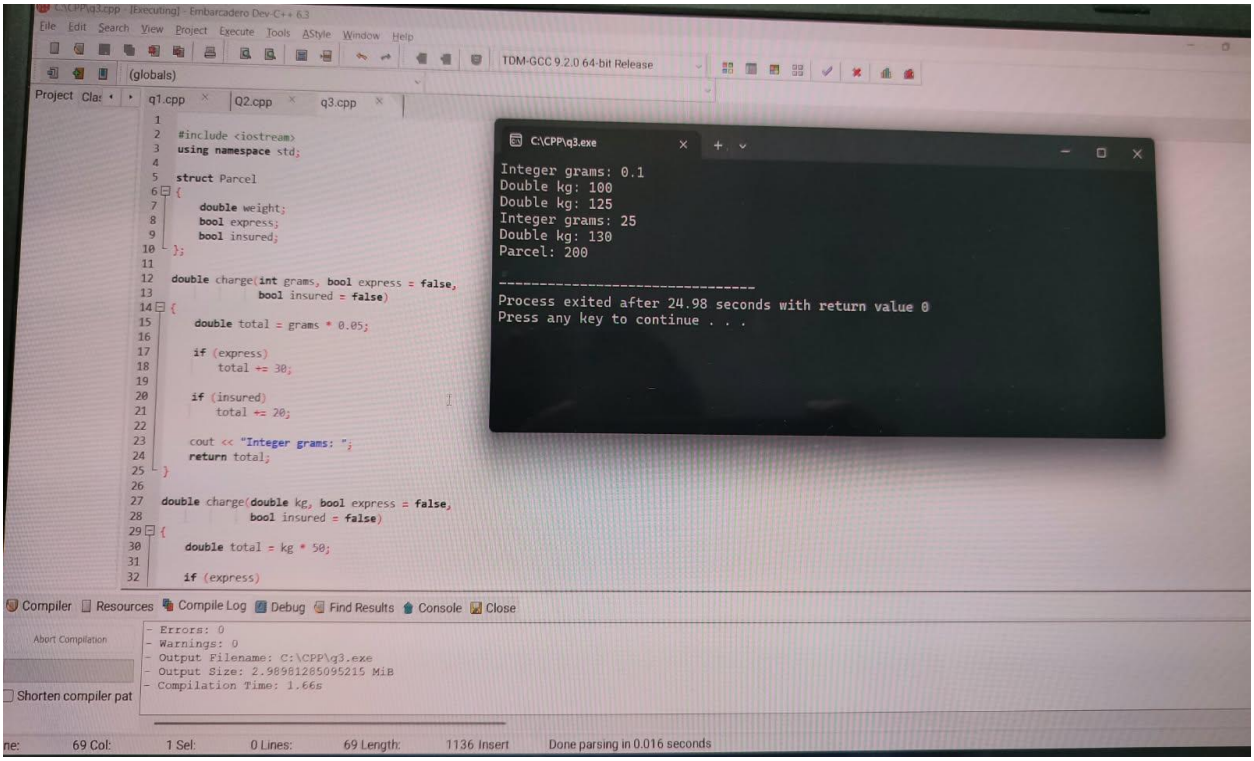
Q1


TDM-GCC 9.2.0 64-bit Release
(globals)
q1.cpp
#include <iostream>
using namespace std;
void change(int a, int& b, int* c)
{
a += 10;
b += 10;
*c += 10;
}
int& bigger(int& p, int& q)
{
return (p > q) ? p : q;
}
void show(int x, int y = 5, int z = 10)
{
cout << x + y + z << " ";
}
void applyByValue(long long fee, int percent)
{
fee -= fee * percent / 100;
}
void applyConcession(long long& fee, int percent)
{
fee -= fee * percent / 100;
}
int main()
Resources
Compile Log
Debug
Find Results
Console
Close
- Errors: 0
- Warnings: 0
- Output Filename: C:\CPP\q1.exe
- Output Size: 2.98988437652588 MiB
- Compilation Time: 1.72s
C:\CPP\q1.exe
1 12 13
1 100
16 13 6
After call by value: 45000
After call by reference: 40500
Process exited after 21.21 seconds with return value 0
Press any key to continue . . .

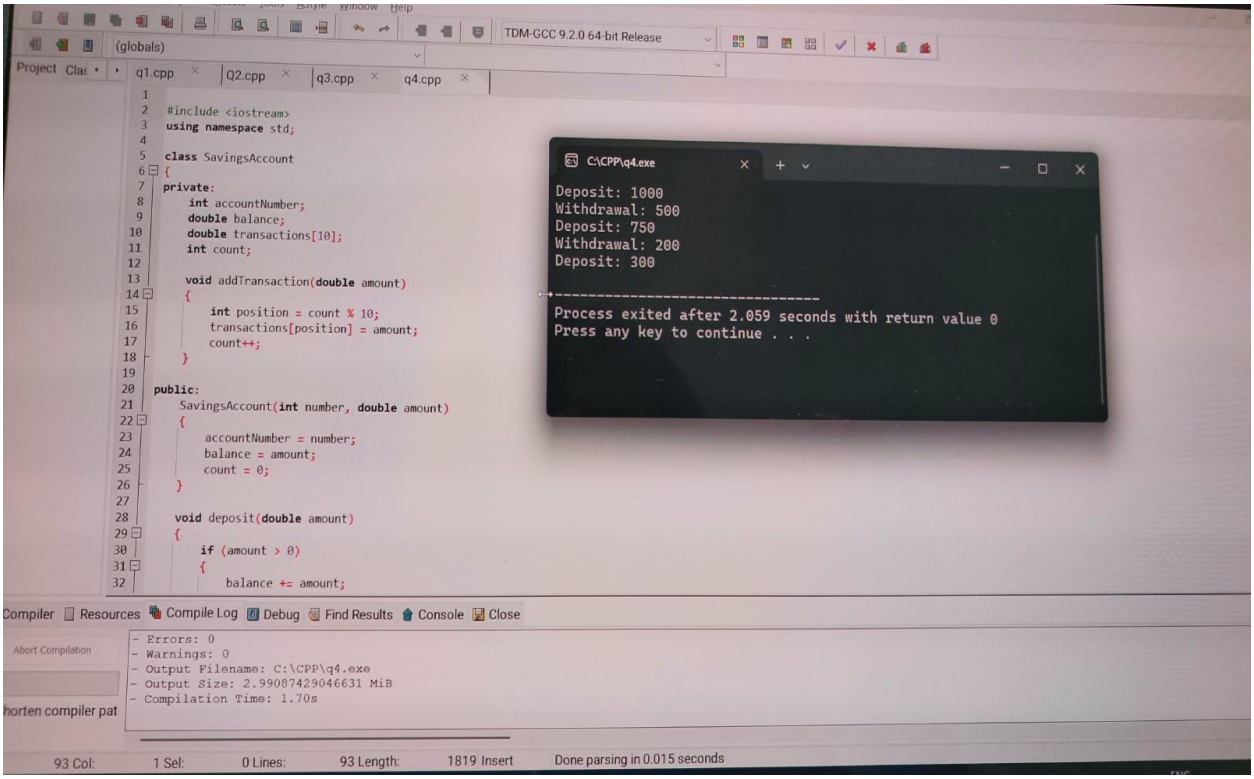
Q2


TDM-GCC 9.2.0 64-bit Release
(globals)
q1.cpp
Q2.cpp
#include <iostream>
using namespace std;
int& highestMark(int marks[], int size)
{
int index = 0;
for (int i = 1; i < size; i++)
{
if (marks[i] > marks[index])
index = i;
}
return marks[index];
}
int main()
{
int marks[5] = {78, 92, 85, 88, 90};
cout << "Highest mark: "
<< highestMark(marks, 5) << endl;
highestMark(marks, 5) += 2;
cout << "After adding 2: ";
for (int i = 0; i < 5; i++)
{
cout << marks[i] << " ";
}
Compiler
Resources
Compile Log
Debug
Find Results
Console
- Errors: 0
- Warnings: 0
- Output Filename: C:\CPP\Q2.exe
- Output Size: 2.98927116394043 MiB
- Compilation Time: 1.78s
Shorten compiler pat
C:\CPP\Q2.exe
Highest mark: 92
After adding 2: 78 94 85 88 90
Process exited after 20.79 seconds with return value 0
Press any key to continue . . .
Done parsing in 0.015 seconds

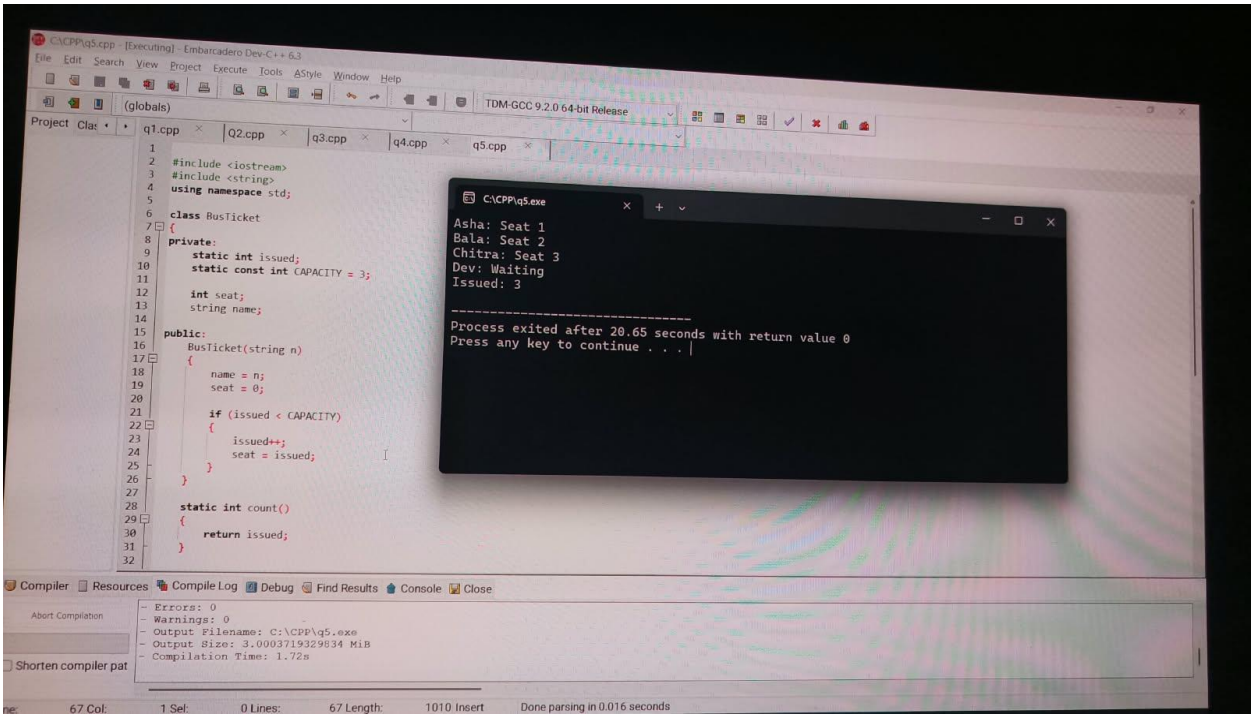
Q3


C:\CPP\q3.cpp [Executing] - Embarcadero Dev-C++ 6.3
File Edit Search View Project Execute Tools AStyle Window Help
TDM-GCC 9.2.0 64-bit Release
(globals)
Project Clas
q1.cpp
Q2.cpp
q3.cpp
#include <iostream>
using namespace std;
struct Parcel
{
double weight;
bool express;
bool insured;
};
double charge(int grams, bool express = false,
bool insured = false)
{
double total = grams * 0.05;
if (express)
total += 30;
if (insured)
total += 20;
cout << "Integer grams: ";
return total;
}
double charge(double kg, bool express = false,
bool insured = false)
{
double total = kg * 50;
if (express)
C:\CPP\q3.exe
Integer grams: 0.1
Double kg: 100
Double kg: 125
Integer grams: 25
Double kg: 130
Parcel: 200
Process exited after 24.98 seconds with return value 0
Press any key to continue . . .
Compiler Resources Compile Log Debug Find Results Console Close
Abort Compilation
Shorten compiler pat
- Errors: 0
- Warnings: 0
- Output Filename: C:\CPP\q3.exe
- Output Size: 2.98981285095215 MiB
- Compilation Time: 1.66s
69 Col: 1 Sel: 0 Lines: 69 Length: 1136 Insert Done parsing in 0.016 seconds

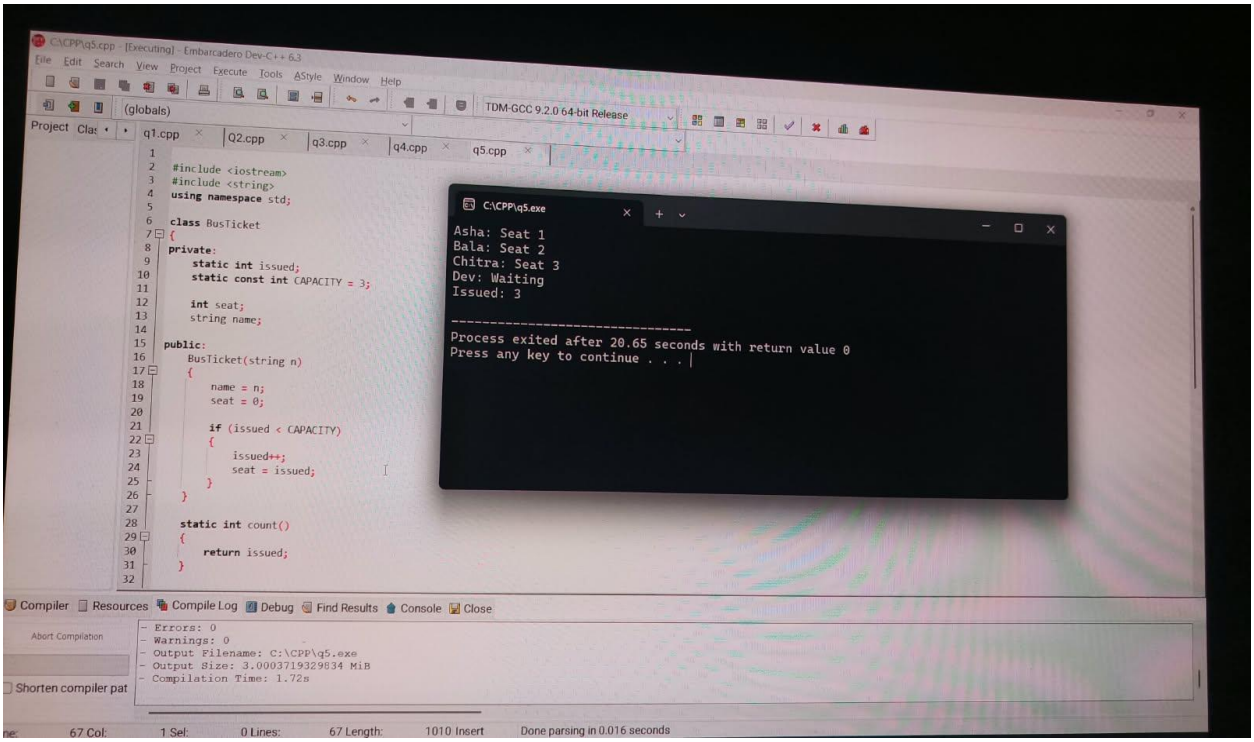
Q4


TDM-GCC 9.2.0 64-bit Release
(globals)
q1.cpp
Q2.cpp
q3.cpp
q4.cpp
#include <iostream>
using namespace std;
class SavingsAccount
{
private:
int accountNumber;
double balance;
double transactions[10];
int count;
void addTransaction(double amount)
{
int position = count % 10;
transactions[position] = amount;
count++;
}
public:
SavingsAccount(int number, double amount)
{
accountNumber = number;
balance = amount;
count = 0;
}
void deposit(double amount)
{
if (amount > 0)
{
balance += amount;
C:\CPP\q4.exe
Deposit: 1000
Withdrawal: 500
Deposit: 750
Withdrawal: 200
Deposit: 300
Process exited after 2.059 seconds with return value 0
Press any key to continue . . .
Compiler
Resources
Compile Log
Debug
Find Results
Console
Close
Abort Compilation
- Errors: 0
- Warnings: 0
- Output Filename: C:\CPP\q4.exe
- Output Size: 2.99087429046631 MiB
- Compilation Time: 1.70s
93 Col:
1 Sel:
0 Lines:
93 Length:
1819 Insert
Done parsing in 0.015 seconds

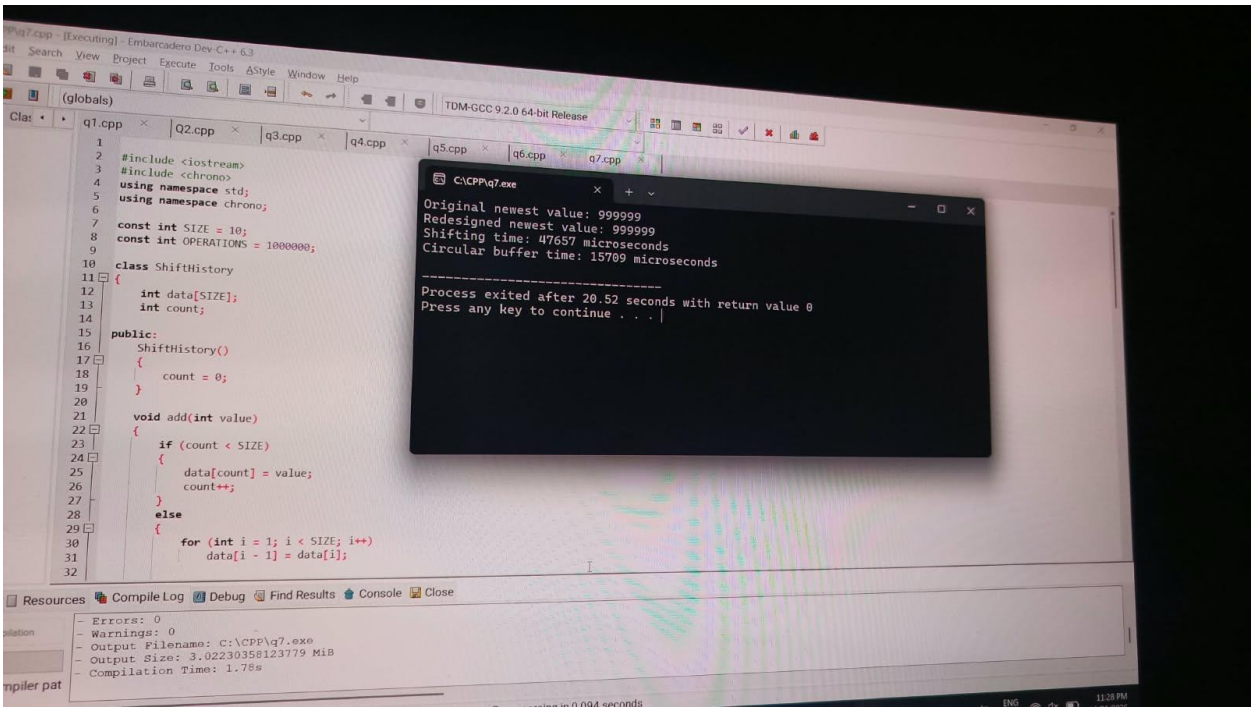
Q5


C:\CPP\q5.cpp - [Executing] - Embarcadero Dev-C++ 6.3
TDM-GCC 9.2.0 64-bit Release
(globals)
q1.cpp
Q2.cpp
q3.cpp
q4.cpp
q5.cpp
#include <iostream>
#include <string>
using namespace std;
class BusTicket
{
private:
static int issued;
static const int CAPACITY = 3;
int seat;
string name;
public:
BusTicket(string n)
{
name = n;
seat = 0;
if (issued < CAPACITY)
{
issued++;
seat = issued;
}
}
static int count()
{
return issued;
}
C:\CPP\q5.exe
Asha: Seat 1
Bala: Seat 2
Chitra: Seat 3
Dev: Waiting
Issued: 3
Process exited after 20.65 seconds with return value 0
Press any key to continue . . .
Compiler
Resources
Compile Log
Debug
Find Results
Console
Close
Abort Compilation
- Errors: 0
- Warnings: 0
- Output Filename: C:\CPP\q5.exe
- Output Size: 3.0003719329834 MiB
- Compilation Time: 1.72s
67 Col:
1 Sel:
0 Lines:
67 Length:
1010 Insert
Done parsing in 0.016 seconds

Q6


C:\CPP\q5.cpp - [Executing] - Embarcadero Dev-C++ 6.3
File Edit Search View Project Execute Tools AStyle Window Help
TDM-GCC 9.2.0 64-bit Release
(globals)
q1.cpp
Q2.cpp
q3.cpp
q4.cpp
q5.cpp
#include <iostream>
#include <string>
using namespace std;
class BusTicket
{
private:
static int issued;
static const int CAPACITY = 3;
int seat;
string name;
public:
BusTicket(string n)
{
name = n;
seat = 0;
if (issued < CAPACITY)
{
issued++;
seat = issued;
}
}
static int count()
{
return issued;
}
C:\CPP\q5.exe
Asha: Seat 1
Bala: Seat 2
Chitra: Seat 3
Dev: Waiting
Issued: 3
Process exited after 20.65 seconds with return value 0
Press any key to continue . . .
Compiler Resources Compile Log Debug Find Results Console Close
Abort Compilation
Shorten compiler pat
- Errors: 0
- Warnings: 0
- Output Filename: C:\CPP\q5.exe
- Output Size: 3.0003719329834 MiB
- Compilation Time: 1.72s
67 Col: 1 Sel: 0 Lines: 67 Length: 1010 Insert
Done parsing in 0.016 seconds

Q7


Search View Project Execute Tools AStyle Window Help
(globals)
TDM-GCC 9.2.0 64-bit Release
q1.cpp
Q2.cpp
q3.cpp
q4.cpp
q5.cpp
q6.cpp
q7.cpp
#include <iostream>
#include <chrono>
using namespace std;
using namespace chrono;
const int SIZE = 10;
const int OPERATIONS = 1000000;
class ShiftHistory
{
int data[SIZE];
int count;
public:
ShiftHistory()
{
count = 0;
}
void add(int value)
{
if (count < SIZE)
{
data[count] = value;
count++;
}
else
{
for (int i = 1; i < SIZE; i++)
data[i - 1] = data[i];
C:\CPP\q7.exe
Original newest value: 999999
Redesigned newest value: 999999
Shifting time: 47657 microseconds
Circular buffer time: 15709 microseconds
Process exited after 20.52 seconds with return value 0
Press any key to continue . . .
Resources
Compile Log
Debug
Find Results
Console
Close
- Errors: 0
- Warnings: 0
- Output Filename: C:\CPP\q7.exe
- Output Size: 3.02230358123779 MiB
- Compilation Time: 1.78s